

Interpreting Diffusion Models Fine-Tuned via DreamBooth Using a Diffusion Lens Approach

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1 Project Overview

We propose to develop an explainable and interpretable AI framework for text-to-image diffusion models by integrating two complementary techniques:

- **DreamBooth** – a method to fine-tune pre-trained diffusion models using a few subject images and a unique identifier (via rare tokens) to enable personalized, subject-driven generation [1]. For more details, please visit <https://dreambooth.github.io>.
- **Diffusion Lens** – an intrinsic tool to *peek inside* the text encoder by generating images from its intermediate hidden states, thereby revealing the layer-by-layer construction of image concepts [2]. Additional details can be found at <https://tokeron.github.io/DiffusionLensWeb/>.

Our integrated approach not only yields high-fidelity image synthesis but also provides insights into how textual prompts are gradually transformed into visual representations. By exploring the interplay between rare token insertion (via DreamBooth) and the progressive representation analysis (via Diffusion Lens), we aim to understand how target image styles are embedded into the model and how these modifications affect the overall text encoding and image generation. Moreover, we will study the role of adapter techniques (e.g., LoRA) in altering the model’s internal representations and visual capabilities.

2 Components

2.1 Application Domain

This project targets the domain of creative and personalized image synthesis. Potential applications include digital art, advertising, and content generation, where understanding model decisions and ensuring subject fidelity are critical.

2.2 Machine Learning Model

Our method builds on state-of-the-art text-to-image diffusion models that typically consist of:

- **Text Encoder:** Converts a natural language prompt (e.g., “a dog in space”) into a latent numerical representation.
- **Diffusion Model:** Uses the text embedding along with an initial noise input to generate a corresponding image.

By fine-tuning these models using DreamBooth, we aim to personalize them for specific subjects by injecting target image information into the model via rare tokens. The Diffusion Lens then allows us to analyze how these modifications propagate through the text encoder and ultimately influence the generated image.

2.3 Proposed XAI Method

Our method comprises two key components which are closely connected:

2.3.1 DreamBooth-based Fine-Tuning

DreamBooth fine-tunes a pre-trained diffusion model with a small set (3–5) of images of a subject by binding a unique, rare-token identifier to the subject. The rare token, chosen from the text encoder’s vocabulary, acts as a minimal prior and is used to encode target image features and styles. The main training objective is given by:

$$\mathcal{L} = \mathbb{E}_{x,c,\epsilon,t} \left[w_t \|\hat{x}_\theta(\alpha_t x + \sigma_t \epsilon, c) - x\|_2^2 \right], \quad (1)$$

where:

- x is the ground-truth image,
- c is the conditioning vector generated from the text prompt (which includes the rare token),
- $\epsilon \sim \mathcal{N}(0, I)$,
- α_t , σ_t , and w_t are parameters controlling the noise schedule.

An additional *prior preservation loss* ensures that the model retains its generalization over the subject’s class even after fine-tuning.

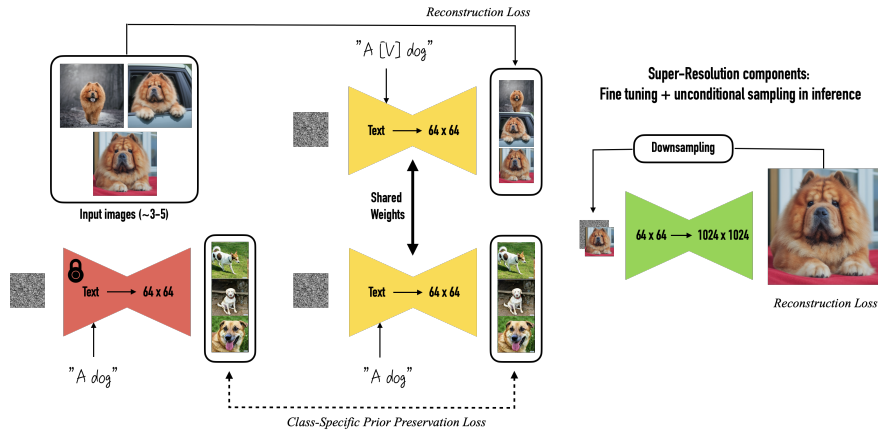


Figure 1: Illustration of the DreamBooth approach: Fine-tuning the diffusion model using rare tokens to encode target subject details and style.

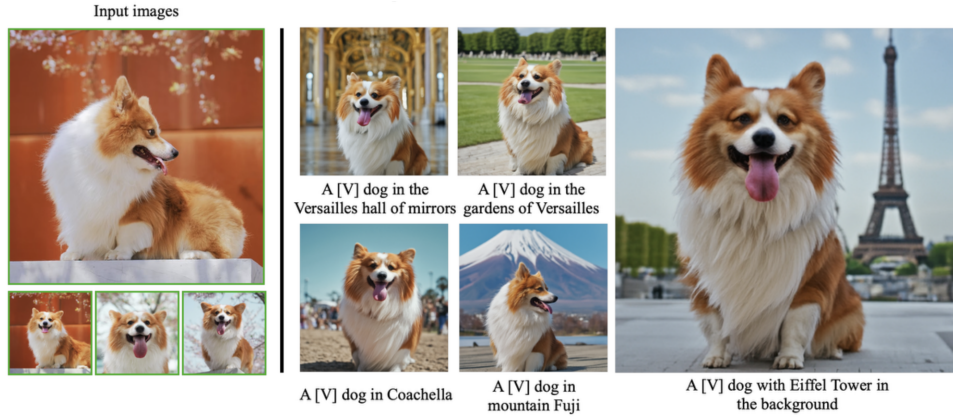


Figure 2: Expected output from DreamBooth fine-tuning: Images generated that exhibit the target subject details and stylistic features as encoded by the rare tokens.

2.3.2 Diffusion Lens for Interpretability

Diffusion Lens is employed to analyze the internal representations of the text encoder after the fine-tuning process. Rather than solely relying on the final output, we generate images from intermediate hidden states. For a given layer l (with $l < L$ for a total of L layers), the generated image is:

$$I = \text{Diff}(\ln_f(h_l)), \quad (2)$$

where:

- h_l is the hidden state at layer l ,
- $\ln_f$ denotes the final layer normalization.

This method provides:

- **Layer-by-Layer Understanding:** Early layers capture basic, unstructured representations (a “bag of concepts”), while later layers progressively refine and organize these ideas.
- **Complexity Analysis:** Simple prompts (e.g., “a cat”) yield clear representations in early layers, whereas complex prompts (e.g., “a red car next to a blue bike”) require deeper layers to form accurate relational structures.
- **Concept Frequency Insights:** Common concepts appear early; uncommon or detailed concepts emerge only in higher layers.
- **Impact Analysis:** By comparing the intermediate representations before and after applying adapter techniques (e.g., LoRA), we can study how such modifications alter the text encoder’s understanding and the final image generation.
- **No Extra Training Required:** The analysis leverages the pre-trained model without modifying its architecture.

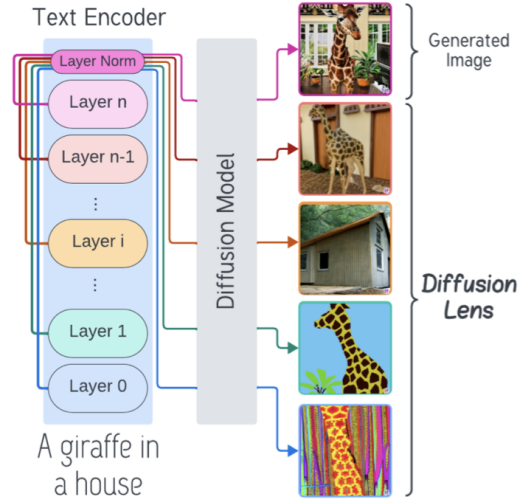


Figure 3: Visualization via Diffusion Lens: Revealing how the text encoder’s intermediate layers build and refine image representations.

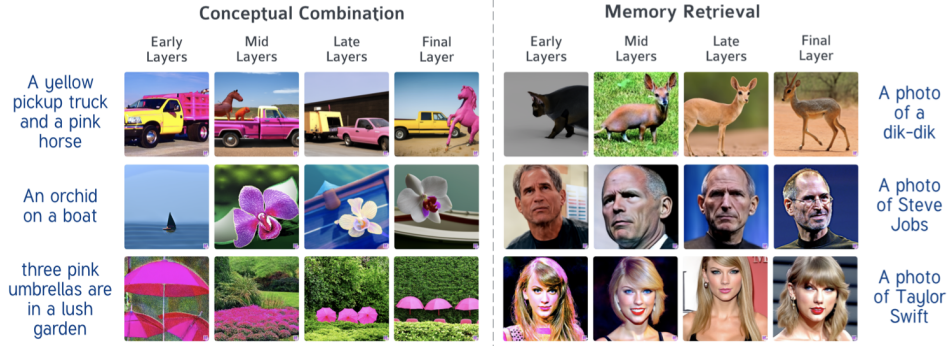


Figure 4: Expected output from Diffusion Lens: A sequence of images generated from intermediate layers, illustrating the gradual refinement of the text encoder’s representation.

2.3.3 Interconnection and Exploration of Adapters

The two components are interlinked:

- **Integration of Rare Tokens:** DreamBooth utilizes rare tokens inserted into the text prompt to encode detailed target subject information. This modification directly influences the text encoder’s intermediate representations, which are then scrutinized using Diffusion Lens.
- **Effect of Adapters:** We will explore how adapters, such as LoRA, modulate the fine-tuning process. The study will focus on assessing changes in the internal representations of the text encoder and the visual output, thereby revealing how such adapters alter both model understanding and visual synthesis capabilities.

3 Deliverables

- **Code Repository:** [Fully documented implementation hosted on GitHub.](#)
- **Technical Blog Post:** A detailed write-up describing the methodology, experimental results, and visualizations.
- **Visual Explanations:** Diagrams, step-by-step image generation sequences, and figures (e.g., Figures 1 and 3) that illustrate the working of both DreamBooth and Diffusion Lens.
- **Evaluation Reports:** Quantitative and qualitative analyses using metrics that assess subject fidelity and prompt alignment.

4 Timeline

1. **Proposal Submission:** March 13, 2025.
2. **Base Implementation:** March 20, 2025.
3. **Experimental Evaluations and Visualizations:** April 15, 2025.
4. **Final Presentation and Blog Post Submission:** April 21, 2025.

5 Conclusion

This project aims to advance explainable AI in text-to-image generation by merging the strengths of Dream-Booth and Diffusion Lens. Through personalized fine-tuning using rare tokens and in-depth, layer-wise analysis, we seek to not only improve image generation quality but also demystify the inner workings of the text encoder. Furthermore, by incorporating adapter techniques, we will explore their effect on model understanding and visual capabilities. The comprehensive visual and quantitative evaluations are expected to yield valuable insights for both research and practical applications.

References

- [1] N. Ruiz, Y. Li, V. Jampani, Y. Pritch, M. Rubinstein, and K. Aberman, *Dreambooth: Fine tuning text-to-image diffusion models for subject-driven generation*, 2023. arXiv: [2208.12242](https://arxiv.org/abs/2208.12242) [cs.CV]. [Online]. Available: <https://arxiv.org/abs/2208.12242>.
- [2] M. Toker, H. Orgad, M. Ventura, D. Arad, and Y. Belinkov, “Diffusion lens: Interpreting text encoders in text-to-image pipelines,” Association for Computational Linguistics, 2024, pp. 9713–9728. DOI: [10.18653/v1/2024.acl-long.524](https://doi.org/10.18653/v1/2024.acl-long.524). [Online]. Available: <http://dx.doi.org/10.18653/v1/2024.acl-long.524>.